

Progress, Challenges, and Prospects of Soft Robotics for Space Applications

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The development of space robots is vital to broadening human cognitive boundaries. Space robots have been deployed in space science experiments, extravehicular operations, and deep space exploration. The application of space robots undoubtedly reduces the risk and cost of space activities. Traditional space robots primarily utilize rigid structures, resulting in limited degrees of freedom, which restricts their operational capabilities. In contrast, soft robots with greater flexibility and robustness may be used for future space exploration. Soft robots applied in space environments must overcome significant challenges associated with ultrahigh vacuum, microgravity, extreme temperatures, and high-energy radiation. Herein, a comprehensive analysis of the key advantages of soft robots is presented based on the special requirements of the space environments for soft robots. Furthermore, brief insights into how soft robots must be changed in terms of their design, modeling, fabrication, sensing, and control to adapt to space environments are discussed. Specifically, soft robot scenarios with potential space application value are introduced. Finally, opinions regarding the potential directions of soft space robots are provided.

relatively mature, the inherently low freedom of rigid robots limits their ability to adapt to space environments and further interactions.^[1] For example, the Spirit rover nearly completely slipped once its wheels became stuck in loose soil.^[2] The high density and complex structure of rigid space robots also increase fabrication difficulty and launch costs.^[3,4] Therefore, to better serve space missions, space robots must have better flexibility and robustness to adapt to complex space environments.^[5]

Soft robots are frequently fabricated from soft materials with Young's modulus of less than 1 GPa.^[6] The unique characteristics of these materials endow them with useful advantages. Generally, compared with rigid robots, soft robots have higher degrees of freedom (DOFs), adaptability, and robustness. Their superior deformability enables soft robots to achieve more dexterous locomotion, which can ensure more

1. Introduction

Space exploration is one of the most challenging and meaningful activities in human history as it can broaden the scope of human cognition of the laws of the universe and origins of life. Currently, a new wave of space exploration is underway. The successful exploration activities of Zhurong and Perseverance rover have once again proved the importance of space robot technology in space exploration. In the future, space robots are predicted to play an increasingly irreplaceable role in space missions. Currently, most deployed space robots have rigid structures. Although the fabrication and control of rigid space robots are

effective interactions with unstructured space environments. The low density of soft robots is also useful for reducing launch costs.^[3] Soft materials can absorb collision energy without damage,^[7] which can reduce the impact of space debris and particles on robots. Therefore, the performance of soft robots exceeds that of rigid robots in some aspects, making them potentially suitable for certain space missions. Note that some institutions began to explore the space applications of flexible mechanisms in the mid-20th century.^[8] Research on Echo balloon satellites,^[8] deployable solar arrays,^[9,10] and inflatable space station modules has achieved impressive application effects.^[11,12] These studies have verified the value of soft machines in space applications. In recent decades, with the progress in advanced materials and control technology, a series of soft robots with space application potential, including pop-up Mars rovers, soft radio antennas, and soft satellite grippers, have been developed.^[13–16] Furthermore, soft robots may be more applicable than rigid robots for some specific space missions (Figure 1). For instance, a deployable soft gripper exhibits conformability when capturing irregularly shaped space debris.^[16] An insect-like untethered flapping-wing aircraft may fly autonomously on a planet with low air pressure.^[17–19] In addition, the soft robot has been proven to withstand collisions without damage.^[17,20,21] A soft robot navigating through tip growth can effectively move in granular media,^[13,22] which may provide some inspiration for solving the problem of rover wheels falling into loose soils. Therefore,

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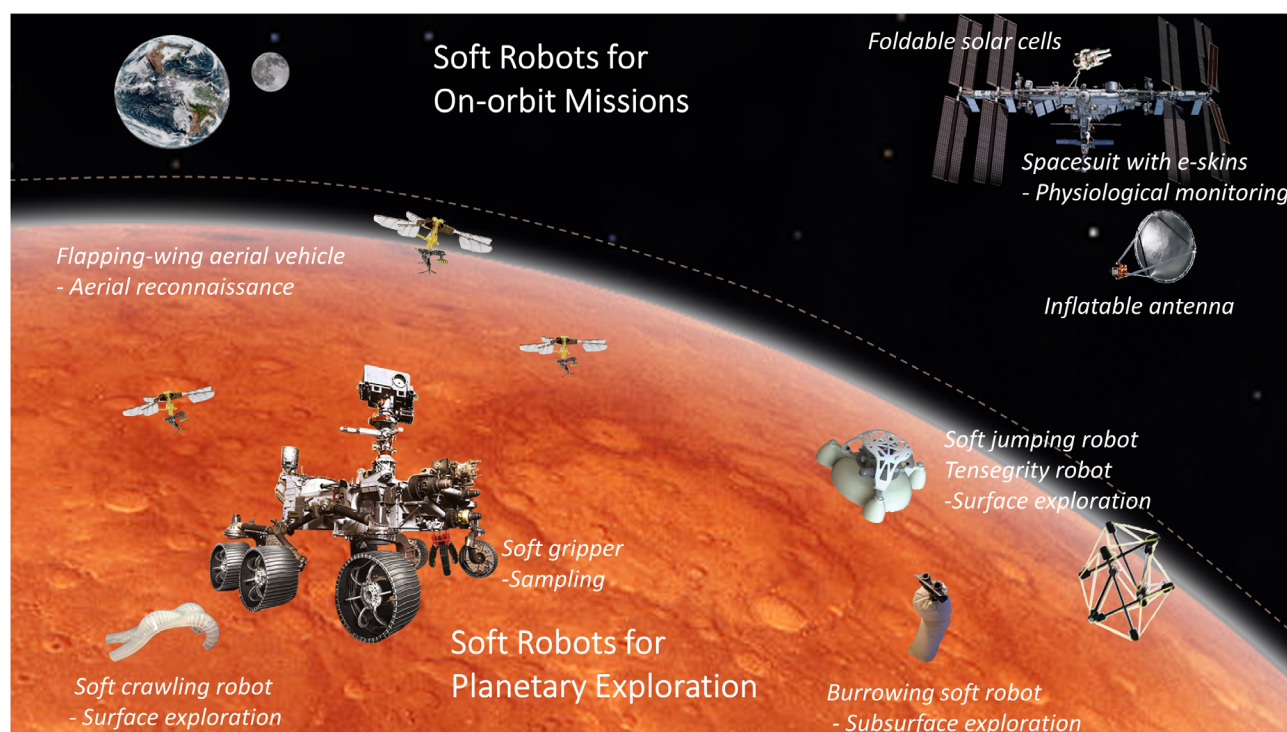


Figure 1. Concept map of space application prospects of soft robots. Soft structures have achieved impressive applications in spacecraft. In the future, soft robots are expected to have a more important role in space equipment such as on-orbit spacecraft, lunar probes, and Mars rovers. In some space missions, soft robots have exhibited potential applications with high adaptability, including origami robots,^[9,23] and kirigami robots^[24,25] for constructing foldable solar cells, soft jumping robots,^[26] soft crawling robots,^[27,28] pop-up rovers for surface exploration,^[15] flapping-wing aerial vehicle for aerial reconnaissance or atmospheric environment detection,^[17] soft grippers for space sampling,^[29] and flexible sensors for astronaut physiological monitoring and human-machine interaction.^[30,31] The soft jumping robot: Reproduced with permission.^[26] Copyright 2015, American Association for the Advancement of Science. The burrowing soft robot: Reproduced with permission.^[22] Copyright 2021, American Association for the Advancement of Science. The flapping-wing aerial vehicle: Reproduced with permission.^[32] Copyright 2021, Springer Nature. The tensegrity robot: Reproduced with permission.^[33] Copyright 2019, American Chemical Society. The soft crawling robot: Reproduced under the terms of the CC-BY-4.0 license.^[28] Copyright 2011, the National Academy of Sciences. The inflatable antenna: Reproduced with permission.^[34] Copyright 2014, American Institute of Aeronautics and Astronautics. The soft gripper: Reproduced under the terms of the CC-BY-4.0 license.^[29] Copyright 2021, the authors, published by Wiley-VCH. The Mars, Mars rover, space station, earth and moon image credit: NASA.

the high flexibility of soft robots provides several new methods to negotiate with highly unstructured space environments.

Nevertheless, most of the current research on soft robotics has focused on terrestrial applications. Research on soft space robots is still in its infancy. Soft robots used in space should satisfy the special requirements of space environments. More specifically, soft space robots must adapt to low-gravity and ultrahigh vacuum environments and have the ability to withstand extreme temperatures and space radiation. However, systematic research in this field is still lacking. In addition, the related design, modeling, and control theories of soft space robots are not sufficiently mature. Therefore, this review focuses on the requirements of space environments for space robots, particularly soft space robots. We summarize significant advancements in soft robots with space application potential. Research progress on soft robots in terms of design, fabrication, actuation, sensing, and control is introduced. Based on these results, the manner in which soft robots should be improved to serve space missions is discussed. Finally, we provide an outlook for future soft space robots. We seek to gain further insights into the evolution of soft space

robots and provide a reference for future development in this field. In addition, it should be noted that the soft robots discussed in this review are soft robots in a general sense, including both soft and flexible robots.

2. Space Environments and Specific Requirements for Soft Robots

Human have developed various space robots for scientific experiments, extravehicular operations, space exploration, and other activities. As a novel type of robot with multiple DOFs, soft robots also exhibit potential for space applications. They are expected to be used for space missions such as on-orbit operations and planetary exploration. However, compared with the Earth's surface environment, the space environment is quite harsh, resulting in more stringent requirements for the normal operation of space robots. The space environment has the characteristics of microgravity, ultrahigh vacuum, extreme temperatures, high-energy particles, and radiation (Figure 2). These physical and

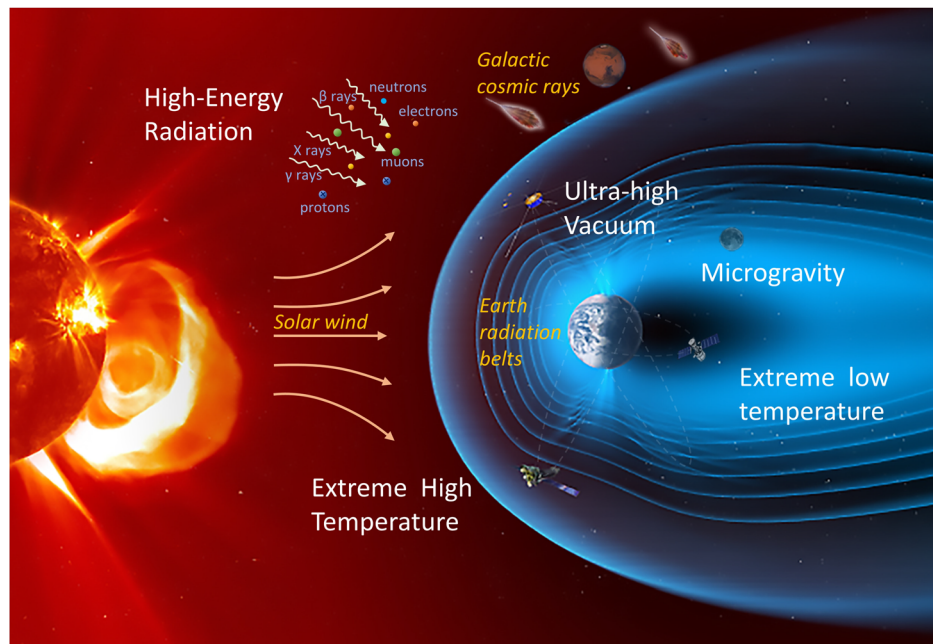


Figure 2. Schematic diagram of space environment characteristics. The space environment is quite harsh compared with the terrestrial environment of humans. It has a series of unique characteristics, including microgravity, ultrahigh vacuum, extreme temperatures, and high-energy radiation. These special physical and chemical characteristics of space environments directly affect the performance of soft space robots. Background image is reproduced with permission from ESA/ATG medialab: <https://sci.esa.int/web/smile/-/61516-cluster-and-xmm-newton-pave-the-way-for-smile>.

chemical characteristics directly affect the performance of space equipment. Therefore, space robots, including soft space robots, must adapt to the characteristics of the space environment and satisfy specific requirements of space applications.

2.1. Characteristics of the Space Environment

2.1.1. Microgravity

The effective gravity of an Earth orbiter is determined by the gravity and inertial centrifugal force generated by its motion. In addition, the orbiter is affected by solar light pressure and gravity gradient effects during operation. Therefore, a spacecraft cannot reach the complete weightlessness state but remains in a microgravity state. The significant gravity difference between the space and ground environment results in different mechanical effects. This causes differences in the motion behavior of space robots between the two environments. Space mechanisms are frequently designed and debugged in ground environments and operated in space microgravity environments. Due to changes in gravity, the modeling and control of space robots are difficult to match in practical applications. In addition, due to a lack of air resistance, even under the action of a small force, space equipment tends to produce large changes in motion states.^[35]

2.1.2. Ultrahigh Vacuum and Extreme Temperature

An important feature of the space environment is its ultrahigh vacuum. Based on the gravity of the Earth, atmospheric pressure gradually decreases with an increase in altitude. It is generally

considered that the outside of the atmosphere (≈ 100 km thick^[36]) is a vacuum environment. Even on the surface of Mars with a thin atmosphere, the large pressure difference poses a significant challenge to the structural strength and sealing of equipment, such as pressure vessels. In addition, due to lack of heat transfer media such as air, space equipment often bears extreme temperatures. Under direct sunlight, the surface temperature of a spacecraft can reach 97 to 127 °C, whereas the temperature decreases to -200 – -150 °C without sunlight.^[37,38] Earth orbiters must be recycled to withstand extremely low and high temperatures. This implies that a spacecraft can only exchange heat through conduction between its components or through radiation between the spacecraft and surrounding environment. The scenario is not optimistic even on the surface of a planet. The ambient temperature on the surface of Mars is ≈ -80 – 0 °C, so warm-up heaters are required on the exterior actuators and cameras of a rover to prevent it from freezing.^[39] Therefore, space robots can experience significant thermal stress.

2.1.3. High-Energy Particles and Radiation

Strong radiation is another important feature of the space environment that differs from the ground environment. Space radiation is generated by charged particles in space. It includes X-rays, β -rays, γ -rays, and high-energy particles such as protons, neutrons, electrons, and muons.^[40,41] Space radiation in Earth's orbit mainly emanates from Earth radiation belts, solar cosmic rays, and galactic cosmic rays.^[42–44] Due to the lack of protection of a geomagnetic field, high-energy radiation can easily bombard the surface and even the interior of space equipment. Therefore,

strong space radiation poses a significant threat to the normal operation of spacecraft and health of astronauts. For example, for semiconductor components in space equipment, the passage of heavy ions in space results in changes in the internal structures of materials, that is, short-term or permanent single-event effects such as disturbance and burning.^[45] In humans, high-energy radiation can cause vomiting, hair loss, and even cancer.^[38,43,46] That is, space radiation has been identified as a significant potential risk to spacecraft and humans. Corresponding protective measures should be considered in the design and fabrication of soft space robots.

2.2. Specific Requirements for Soft Space Robots

A harsh space environment entails stricter requirements for the design, fabrication, actuation, sensing, and control of soft space robots. Soft space robots must adapt to the space environment to ensure normal operation. The complexity of space missions poses significant challenges to the modeling and design of soft space robots. Based on the different compliances of underlying materials, the design and modeling methods of soft robots are significantly different from those of rigid robots.^[6] The inherent strong nonlinearity of soft materials increases modeling complexity. Generally, the modeling accuracy of soft robots is difficult to ensure compared with that of rigid robots. In addition, the design theory of soft robots remains immature. Soft organisms and muscles have long served as important sources of inspiration in the design of soft robots. Researchers have developed several bioinspired automated design algorithms, but universal design theory remains a major challenge.

Compared with the metallic materials that are commonly used in rigid space robots, soft materials are generally less adaptable to harsh space environments, including space radiation and extreme temperatures.^[47] Remarkably, it is difficult to replace or repair damaged devices in a space environment,^[5] so it is necessary to ensure the long-term stability of space devices. Therefore, it is critical to develop soft space materials that can withstand space environments and exhibit long-term stability. Furthermore, the consistent reliability of soft space robots must be ensured through strict online tests and the verification of environmental stability.

Good manufacturing compliance is the basis for the effective operation of soft space robots. In a space environment, microgravity and extreme temperatures may amplify the mechanical differences in the interlayer structures of soft materials, generating cracks. In addition, an ultrahigh vacuum environment requires a soft robot, particularly a fluid-actuated soft robot, to have a higher sealing level. The impact of space debris can also disrupt the structural performance of space robots by generating denudation and cracking.^[37,48] Space radiation and extreme temperatures can easily cause plastic deformation, creep, aging, and even degradation of soft materials.^[49,50] To overcome these challenges, manufacturing methods must be improved. A feasible solution is to develop space robots with multiple protective layers to shield from space radiation and maintain the relative stability of internal states. Therefore, compared with rigid space robots, soft robots require more comprehensive and systematic protection measures.

In the Earth's orbit or on different planets, the physical and chemical characteristics of the environment may vary significantly. Therefore, it is crucial to select an appropriate actuation method for soft space robot in different scenarios. For example, light, heat, or magnetic actuation can be considered for space missions that require remote control. Chemically actuated soft robots may be more suitable for highly acidic or alkaline environments on outer planets. Soft space robots require higher actuation efficiency than ground-based soft robots based on the high cost of space equipment maintenance and restriction of energy supply.^[4,51,52]

Sensing and control methods for soft space robots require further improvements. Soft robots in Earth's orbit float freely in space. In space missions, an actuator typically approaches and interacts with a service target. Therefore, compared with a ground-based soft robot, a soft space robot must control its position and attitude more accurately. The premise of effective control of soft robots is to obtain the key parameters of robots and their environments (e.g., position, attitude, strain, and temperature). Flexible sensors and microsensors can be widely used in the sensing of soft robots because they can be embedded in soft robots without affecting their motion. The main challenge is that it is difficult to achieve precise control of soft robots, even with accurate state information. Considering their inherent flexibility, soft robots with conventional control strategies often exhibit good performance on the ground. However, in a highly unstructured space environment, it is difficult for these control strategies to satisfy space mission requirements.

3. Design and Modeling

Soft robots are lightweight, robust, and have multiple DOFs, resulting in excellent potential for space missions.^[53] However, unlike rigid space robots, design and modeling methods for soft space robots are lacking. Current design tools often cannot adapt well to the complex inhomogeneous design of soft robots due to the difference in the compliance of the underlying materials.^[6] Furthermore, space factors such as microgravity, ultrahigh vacuum, and extreme temperatures impose various internal and external stresses on soft space robots and shorten their lifespan.^[39] High-energy particles and high-speed debris can strike and damage the surfaces of soft robots. All these factors make it difficult to construct a dynamic model that accurately describes a soft robot. The main challenge is that we have not been able to understand the influence of space environments on the locomotion of soft robots systematically. In addition, the strong nonlinear effects of soft materials and structures increase design difficulty.

3.1. Modeling

Unlike rigid robots, soft robots follow amorphous nonrigid dynamics.^[54] Strong nonlinear effects require numerous calculations, making it difficult to guarantee accuracy. In addition, the harsh space environment with ultrahigh vacuum, microgravity, extreme temperature, and high-energy radiation, makes conventional modeling methods inapplicable or necessitates additional corrections. Currently, research on soft robot modeling has

primarily focused on soft material modeling, deformation kinematic modeling, and flexible structure contact dynamics.

Establishing a constitutive model for soft materials is the basis for understanding the kinematics of soft robots. Mooney first proposed a model for incompressible hyperelastic materials such as rubber in 1940.^[55] Subsequently, the Mooney-Rivlin,^[56] Ogden,^[57] Arruda-Boyce,^[58] and Yeoh models have also been proposed.^[59] Suo et al. proposed a constitutive model and large deformation theory for dielectric elastomers (DEs) based on continuum mechanics and thermodynamics.^[60,61] Their model revealed the force–electric stability of DEs under an electric field. DEs are one of the few smart material types that have been used to develop space capture devices.^[16]

Soft robots are multiflexible dynamic systems with geometric nonlinearity and large strains. The kinematics and large-deformation dynamics of soft structures have been studied extensively. Theoretically, the actual motion state of a soft robot can be described by an instantaneous state function. However, this modeling method is limited by current dynamic theory and computing power. It is feasible to simplify the locomotion of a soft robot appropriately. The development of inflatable space robots has demonstrated the feasibility of fluid-actuated soft robots for space applications.^[62] The piecewise constant-curvature (PCC) model is commonly used to describe fluid-actuated soft robots. This model approximates a soft manipulator axis as a series of mutually tangent arcs of constant curvature. Webster et al. distinguished the robot-specific and robot-independent parts of a PCC model using differential kinematics.^[63] Jones et al. analyzed the relationship between joint variables and curvature arc parameters, thereby establishing a model between robot workspace coordinates and actuator inputs.^[64] Experiments demonstrated that their model could describe the deformation of high- and medium-pressure pneumatic soft robots. Renda et al. developed a dynamic model of a cable-actuated soft robotic arm.^[65] Their model can accurately describe the dynamic behavior of a continuum soft robot in dense media. For hyperelastic thin-shell space structures such as flexible solar cells or inflatable cabins, Luo et al. proposed a computational model for nonlinear static analysis.^[66] The coupling effects of large deformations and rotations of hyperelastic thin-shell structures can be fully described using this method. In their study, the dynamic simulation results of an inflatable torus and solar sail were consistent with experimental results. Schenk et al. proposed a simplified hinged-rod truss model for origami structures that enabled efficient kinematic and stiffness analyses during the folding process,^[67] which has the potential to be used in the analysis of collapsible solar arrays.

In a microgravity environment, the contact or collision of soft robots with external objects significantly changes their state. Therefore, establishing an accurate contact dynamics model is beneficial for optimizing design and control. In addition to conventional methods (e.g., the Coulomb and LuGre models),^[68,69] Benvenuto et al. analyzed the process of net deployment, contact, and closure dynamics of a tethered net robot on a target based on multibody dynamics.^[70] The numerical results indicated that tethered net robots are a promising technology for capturing and removing space debris. In addition, forward and inverse kinematics analyses aid in understanding the global motion behavior of soft robots and provide a reference for the modeling of soft robots.^[71,72]

3.2. Design

Lightweight and flexible soft robots have excellent potential for applications in space unstructured environments. The design of soft robots is extremely complicated based on nonlinear materials and multiphysical field coupling. Constructing robots capable of completing complex space missions depends on efficient modeling and simulation tools.^[73] These methods are still lacking for soft robots.

Nature provides a steady source of inspiration for the design of soft robots. Drawing inspiration from natural creatures may simplify the design process and lead to better designs. In the early days of soft robotics research, due to the lack of effective design tools, manual design based on biomimetic methods was one of the main methods. For example, Kim et al. developed a pelican-eel-inspired dual-morphing architecture.^[74] A soft robot based on this architecture can efficiently complete underwater motion for large-scale capture and may also be deployed for space debris capture. Furthermore, various optimization methods have been used in soft robots to achieve local or global automation designs. Connolly et al. developed an analytical strain energy density function method that can automatically design specific parameters according to the deformation of fiber-reinforced pneumatic actuators.^[75] This approach promises to provide quantitative solutions rapidly for a soft robot performing specific space missions, but its further development relies on the experience of designers. In contrast, numerical finite-element analysis (FEA) has stronger versatility without expert knowledge. Chen et al. developed an FEA model to examine the effects of various geometric parameters on the behavior of soft twisting pneumatic actuators.^[76] The complex nonlinear effects between geometric parameters and pressure loads can be modeled effectively. However, FEA also has limitations. Due to the limitations of computing power and material nonlinearity, it is difficult to achieve large design spaces with complex interactions. Topological optimization (TO) works by distributing materials within the design space to find a structure that maximizes a fitness function.^[77] This method can effectively reduce the mass of space equipment without affecting structural strength.^[78] Relying on TO, many novel designs have been generated in recent years. Chen et al. successfully used the TO method to optimize cable-driven soft grippers for grasping specific objects of different shapes and weights.^[79]

In addition, evolutionary algorithms can efficiently search the design space to obtain high-performance designs for soft robots.^[80–82] Sims was the first to demonstrate that computational evolution using genetic algorithms can simulate locomotion similar to that of natural organisms.^[83] Subsequently, Cheney et al. developed a compositional pattern-producing network (CPPN) algorithm that can generate the shape of a robot by specifying the material type of the spatial coordinate function.^[84] Hiller et al. applied a spatial evolutionary algorithm to the automatic design of freeform soft robots. The performance error of a robot fabricated using this algorithm was confirmed to be less than 15%.^[85] In the study of the complex behaviors of soft robots, evolutionary robots frequently outperform traditional designs. However, such robots are rarely fabricated as a result of the

complexity of the real environment relative to the simulated environment.

Advanced design methods can ensure that soft robots are lighter and stronger, thereby meeting the requirements of aerospace engineering. Furthermore, considering the characteristics of space missions, specific designs are indispensable, including strict protective layer design. In addition, soft robots are typically unsuitable for sustaining strong loads, which may limit their space application. Relative to the ground environment, moving the same mass in microgravity requires only a small force.^[33] However, the low stiffness of soft robots can still lead to the failure of space missions, such as the escape of captured space debris.^[86] Antagonistic variable stiffness, phase transition variable stiffness, and jamming structures are several commonly used methods to achieve tunable mechanical behavior in soft robots.^[87–89] In contrast, jamming structures are more likely

to be used in soft space robots because of scalability, easy fabrication, and low cost. Soft robots with the ability to actively control stiffness allow for adaptable and versatile interactions between robots and space environments. Adjustable mechanical properties enable soft space robots to take into account high precision, high force, compliance, and safe interactions. In summary, we provide a brief chronology of soft robot modeling and design methods in Figure 3.

4. Materials and Fabrication

4.1. Soft Materials for Space Application

The durability of materials in space environments plays an important role in the long-term fault-free operation of space

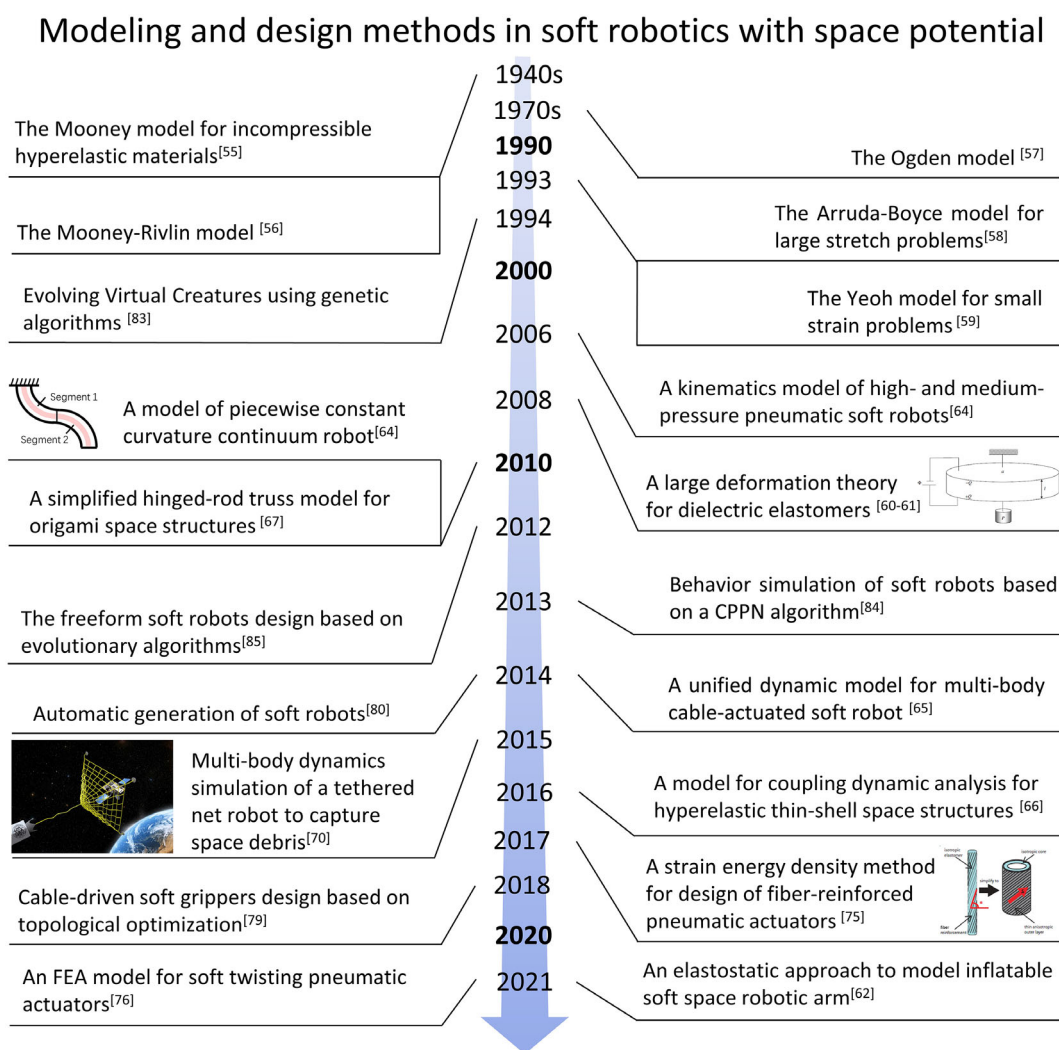


Figure 3. A brief chronology of the modeling and design methods of soft robots with space applications potential. We emphasize the typical modeling and design methods for soft structures and robots. These methods have important applications in soft robot design and have significantly promoted key technological progress in some respects. The diagram for a large deformation theory of dielectric elastomers: Reproduced with permission.^[60] Copyright 2008, Elsevier. The schematic diagram of a tethered net robot: Reproduced with permission.^[70] Copyright 2015, Elsevier. The deformation diagram of a fiber-reinforced pneumatic actuator: Reproduced under the terms of the CC-BY-4.0 license.^[75] Copyright 2017, the National Academy of Sciences.

robots. Soft robots are typically composed of soft materials with Young's modulus less than 1 GPa.^[6,53,90] These materials are the basis for the fabrication, actuation, and control of soft robots. However, the soft materials used in conventional soft robots are often difficult to adapt to unstructured environments. A harsh space environment can easily cause material aging and structural failure. For example, ordinary rubber tends to become brittle and crack at extreme low temperatures. Changes in material properties caused by space factors determine the service states of the corresponding space structures. In current aerospace engineering, materials that balance adaptability, functionality, and self-healing are lacking. Therefore, it is critical to design or select suitable materials for space applications. Currently, design-based solutions rely on the use of redundant and doubled systems, leading to a decrease in carrying capacity. In addition to adapting to various extreme space conditions, materials for future soft space robots should have the ability for active self-healing to ensure the long-term reliable operation of systems. In **Table 1**, we list existing and potential materials for space applications and specifically mark potential soft materials for manufacturing soft space robots. Fiber composites and hyperelastic self-healing alloys, which are commonly used in current aerospace engineering, may play a role in future soft space robots. Furthermore, to realize the long-term reliable operation of soft space robots, soft space materials should consider adaptivity, functionality, and self-healing.^[91] Potential soft materials, such as liquid metal, temperature-resistant elastomers, and smart nanostructured protective coatings, offer several feasible options for future soft space robots.^[92–95] These materials can be used as prime candidates for elastic inflatable actuators, inflatable antennas, and soft space robots.^[11,96,97]

4.2. Fabrication

Currently, it is feasible to construct a soft robot on the ground and launch it into the space environment. However, in the long run, on-orbit manufacturing and even in situ manufacturing are very promising. Considering the harsh space environment, the fabrication of soft space robots is far more complicated than that of conventional soft robots. The current fabrication techniques for soft robots remain the basis for manufacturing soft space robots but must be improved. The most widely used fabrication techniques include 3D/4D printing,^[112] injection molding,^[113] lamination,^[15,114] soft lithography,^[28,115] and transfer printing.^[116] (**Figure 4**). **Table 2** lists the typical parameters and scenarios applicable to these techniques.

4.2.1. 3D/4D Printing

Additive manufacturing (also known as 3D printing) can rapidly create 3D objects by continuously adding soft material layers via rotational casting, direct writing, and screen printing. The production of parts in the space industry is small scale and highly customized.^[117] 3D printing techniques could be quite helpful for the fabrication of custom mission-adapted soft space robots based on the rapid and cheap adaptation of the designs.^[118] These techniques have been used to manufacture soft sensors,^[119,120] soft actuators,^[121] and even integrated soft robot systems.^[26] By precisely controlling the relative stiffness of each local unit, 3D-printed soft robots can achieve programmable deformation.^[122–124] 3D printing techniques that can be used to manufacture soft robots include fused deposition modeling (FDM), direct ink writing (DIW), selective laser sintering (SLS), inkjet printing (IP), and stereolithography (SLA).

Table 1. Existing and potential materials for space applications.

Special requirements for space materials	Existing materials	Potential materials
Temperature tolerance	Metal materials (titanium alloy) Fiber materials (Kevlar) ^{a)} Organic materials (polyimide, teflon, etc.) ^{a)} Inorganic materials (Si solar cells, ceramics)	Liquid metal ^{[98]a)} SolarBlack material ^[99] VO ₂ -based smart coatings ^{[94]a)} Temperature-resistant elastomers ^{[93]a)}
Radioresistance	UV-inert film (Kapton, aluminized mylar) Lead-based materials Si cells, gallium arsenide solar cells Low-temperature curing epoxy resin ^[100]	Zr/Nb nanoscale multilayer coatings ^{[101]a)} Nanocrystalline diamonds ^{[91]a)} In-site regolith composites ^{[100]a)} Nanostructure arrays ^{[102]a)}
Anticorrosion	Fiber materials (Kevlar) ^{a)} Structured ceramics ^[91] chromate (CrVI-based) conversion coatings ^[103]	Anticorrosion coatings based on pH-sensitive polyelectrolyte nanostructures ^{[95]a)} Hexavalent chromium-free anticorrosion coatings ^[103]
High strength	Metal matrix composites ^[91] Ceramics matrix composites ^[91] Polymer matrix composites ^{[91]a)}	Fibers made of carbon nanotubes ^{[104]a)} Graphene-reinforced composites ^{[91]a)} Carbon filament-reinforced composites ^{[91]a)}
Self-healing	Fe–Au alloys ^[105] Supersaturated solid solution materials ^{[106]a)} Metallic matrices reinforced with shape memory alloys ^[105] Reversible-forming crosslinkers ^{[91]a)} Metallic matrices reinforced with low-melting alloy ^[106]	Graphene-based smart materials ^{[107]a)} Self-healing alloys ^[108] Nanomotors of conductive particles ^{[109]a)} Self-healing elastomer (hydrogel/aerogel) ^{[110]a)} Cu/V nanolayer ^{[111]a)}

^{a)}denotes soft materials that may be used to manufacture soft space robots.

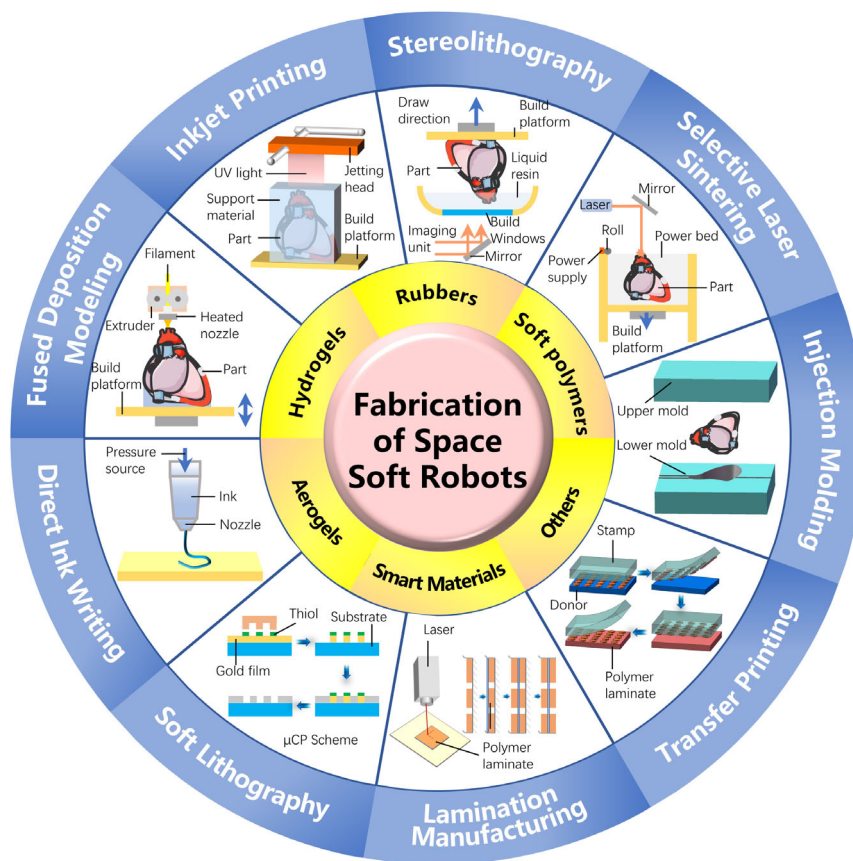


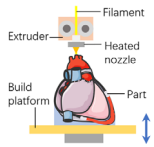
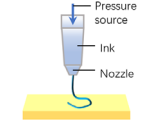
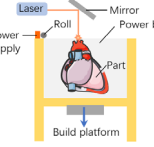
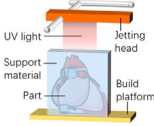
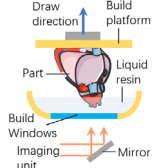
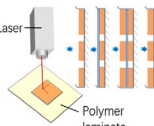
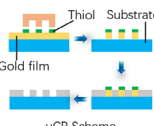
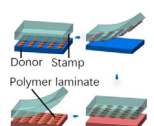
Figure 4. Common technologies for fabricating soft space robots. Common technologies for fabricating soft robots include 3D/4D printing, injection molding, transfer printing, lamination manufacturing, and soft lithography. 3D/4D printing can be further divided into FDM, DIW, SLS, IP, and SLA.

Compared with 3D printing, 4D printing has an extra time dimension, meaning that it can fabricate structures that are time deformed using stimuli.

Up to now, most 3D-printed aerospace components have been made of metal materials, as usually they are stronger and more flame retardant than soft materials.^[125] Recently, soft materials have attracted interest in some institutes because of their fuel efficiency. 3D-printed soft polymer composites that can withstand high temperatures have been used in space applications. Based on the FDM process, a rocket engine inlet guide vane manufactured by the Glenn Research Center can withstand temperatures exceeding 200 °C.^[126] Wu et al. developed layered composite structures based on shape memory polymer (SMP) fibers using 4D printing.^[127] Following simple thermomechanical programming, complex deformation behaviors, such as bending and helix shape, can be achieved under different temperature fields. Diverse deformability makes it possible to create soft robots adapted to different space missions. In addition, 3D printing presents an opportunity for manufacturing soft space robots with integrated sensing. Truby et al. developed a soft actuator integrating haptic, thermal, and proprioceptive sensing based on improved embedded 3D printing.^[128] This may be applied in closed-loop control of soft robots, space wearable devices.

In-space 3D printing is a powerful method for improving reliability and reducing launch costs. However, limited by microgravity, ultrahigh vacuum, space radiation, and other conditions, the efficiency of in-space 3D printing may be reduced. Soft materials, such as rubber and plastic, are easy to age under the effects of space radiation. Extreme temperatures may cause materials extruded by an in-space 3D printer to have difficulty curing or become brittle.^[129] Therefore, it is necessary to package the printer carefully to maintain the relative stability of its internal temperature. Yirmibeşoğlu et al. verified that the mechanical properties of polydimethylsiloxane (PDMS) decreased with increasing exposure to gamma irradiation.^[130] Under gamma radiation up to 20 kGy, 3D-printed soft robots fabricated from PDMS can only operate for ≈12 h. Some soft printable materials have been proven to withstand ground-simulated space environments, such as cosmos radiation, thermocycling, atomic oxygen, and high vacuum.^[53] For instance, polypyrrole-based soft robots have potential as devices for vacuum applications.^[131] In another study, Darabi et al. revealed the 3D printability of such materials.^[132] In-space 3D printing also has some natural advantages. For example, in a microgravity environment, redundant support structures are no longer required, which improves material utilization and reduces the overall mass of structures.

Table 2. Typical fabrication methods for soft robots and their space application potential.

Parameter	Fabrication method	Fabrication efficiency	Approximate resolution	Material requirements	Applicable scenarios and space application potential
FDM		$10^5 \text{ mm}^3 \text{ h}^{-1}$ ^[143] Inversely proportional to resolution	$100 \mu\text{m}$ ^[143]	Thermoplastic (filaments, such as PLA, ABS, peek, nylon)	In-cabin manufacturing (Soft, flexible structures of soft space robot)
DIW		$10^{-3}-10^5 \text{ mm}^3 \text{ h}^{-1}$ ^[144] Inversely proportional to resolution	$1-100 \mu\text{m}$ ^[145]	Flowable (ink, hydrogel, liquid metal ^[92]) that solidifies	In-cabin, in-suit manufacturing ^[146] (Soft structures of soft space robot)
SLS		$10^3-10^6 \text{ mm}^3 \text{ h}^{-1}$ ^[144]	$5-100 \mu\text{m}$ ^[147]	Thermoplastic (powder, polycaprolactone, the planet soil, Moon and Mars regolith)	In-cabin, extravehicular, in-situ manufacturing (Rigid, flexible structures of soft space robot)
IP		$10^4-5 \times 10^5 \text{ mm}^3 \text{ h}^{-1}$ ^[144]	$50 \mu\text{m}$ ^[148]	Low viscosity (ink), powder $V_{\text{app}} < 0.25 \text{ Pa s}$ ^[148]	In-cabin, extravehicular manufacturing (Soft, flexible structures of soft space robot)
SLA		$10^3-10^6 \text{ mm}^3 \text{ h}^{-1}$ ^[144] Slower rates for microsystems	100 nm ^[149] (microsystems) $50 \mu\text{m}$ ^[150] (projection-based)	Low viscosity (ink, epoxy, hydrogel,) $V_{\text{app}} < 5 \text{ Pa s}$ ^[151] photopolymerizable	In-cabin (Soft, flexible structures of soft space robot)
Lamination manufacturing		—	$5 \mu\text{m}$ ^[114]	Metal plates, polymer films, fiber composites, etc.	In-cabin, extravehicular manufacturing (Mars, Lunar rover, etc. soft, flexible structures of soft space robot)
Soft lithography		10–20 s to form ordered self-assembled monolayers ^[115]	$0.03-100 \text{ nm}$ ^[115]	Desensitized polymers, sol-gel materials, conducting polymers, etc.	In-cabin manufacturing (Soft, flexible structures of soft space robot)
Transfer Printing		$0.02-4.8 \text{ cm s}^{-1}$ ^[116] (Separation) $\geq 10 \text{ cm s}^{-1}$ ^[116] (peeling)	100 nm ^[116]	Silicon wafer, GaN ribbons, MgO substrate, etc.	In-cabin manufacturing (Protective layers, soft, flexible structure of soft space robot)

4.2.2. Lamination Manufacturing

Lamination manufacturing, which is also known as the smart composite microstructure manufacturing process, is another method for the rapid manufacturing of soft robots. Laminate structures were inspired by pop-up books. This process is suitable for fabricating a flexible articulated 3D structure with features on the order of micrometers to centimeters.^[15,114] Lamination manufacturing enables the integration of a rigid link fabricated from fiber composites with flexure joints fabricated from polymer films.^[90,133] This provides a method for electronic

devices to be integrated directly into soft robots. Karras et al. developed an origami-inspired pop-up robot for future space exploration, particularly for Mars missions.^[15] A textile-enhanced rigid-flex printed circuit was laminated to obtain the robot a chassis. This chassis can realize structural variants through self-folding to adapt to various terrains. In addition, lamination manufacturing has also been used in the fabrication of scalable articulated robots with integrated sensing capabilities,^[134] and self-folding robots,^[135] which are very useful for developing morphing aerospace mechanisms.^[136] For example, as one of the main components in satellites supporting the tip payloads,

traditional gravity gradient booms are mainly actuated by motors, which are heavy and of high cost. Zhang et al. developed an SMP composite truss boom consisting of multiple pieces of laminate tape, which can be gradually deployed within 80 s.^[137]

4.2.3. Injection Molding

Injection molding is a simple, efficient, and low-cost method for manufacturing soft robots. Injection molds are typically obtained through machining or 3D printing. To obtain injection molds, the required structural entities can be created through raw material mixing, injection molding, demolding, and other processes.^[113] Based on the ultrahigh vacuum conditions in space, fluid-actuated soft space robots fabricated using injection molding often require additional processes such as vacuum deaeration and fiber enhancement.

In addition to the earlier methods, transfer printing,^[116,138] soft lithography,^[28,139] and shape deposition^[140] are commonly used for the fabrication of soft robot systems. Some significant challenges must be addressed to obtain a practical soft space robot. For example, strong space radiation can cause the failure of materials and electronic components.^[37] An effective solution involves covering the surfaces of soft robots with multilayer insulation or coatings.^[141,142] Generally, a protective layer is fabricated from aluminized polyester films, polyester fiber, etc., which can isolate heat and substance exchange within soft robots. However, this method may affect the movement of soft robots. Further research is important for determining how to alter the fabrication process to make soft robots more suitable for space applications.

5. Actuation

The apparent differences between space and Earth surface environments require the actuation of soft robots to change accordingly. For example, the space ultrahigh vacuum environment poses a significant challenge to the application of pneumatic soft robots. Therefore, soft robots must adapt to these changes in space exploration. There are two main types of actuation modes for soft robots: physical and chemical. Depending on the actuation source, they can be divided into motor, fluid, smart material, chemical actuation, etc. Motor actuation is not described in detail here as it is the main actuation form of traditional space robots.

5.1. Physical Actuation

5.1.1. Fluid Actuation

Fluid actuation outputs force use hydrostatic pressure, which enables soft robots to produce sophisticated motion under simple control.^[28] It has a wide range of adaptability. A tip-growth soft robot for navigation and detection,^[13,22] pneumatic soft robots for running (Figure 5a)^[152] or crawling,^[27,28] and hydraulic soft robots for swimming^[153] or capturing have been developed.^[154] Typically, fluid-actuated soft robots must carry hydraulic or pneumatic systems, which may limit their locomotion efficiency. Furthermore, fluid-actuated soft robots are highly

sensitive to ambient temperature, which can easily affect their performance. For instance, the Solar Orbiter can withstand temperatures as high as 520 °C while performing in situ measurements of the Sun.^[155] The temperature of PDMS, which commonly used in soft robots, should not exceed 200 °C. Otherwise, it can be thermally decomposed.^[156] When the Sun is blocked by the Earth, the surface temperature of the International Space Station decreases to ≈ -160 °C.^[37] Extremely low temperatures can freeze the fluids in soft robots. In addition, another study demonstrated that the degradation of silicone rubber is accelerated under the action of cycling temperature.^[157] Therefore, it is critical to control the temperature of fluid-actuated soft robots to be relatively stable. Another solution is the development of soft materials that can withstand extreme temperatures. For example, the thermal stability of silicone can be significantly increased by removing the sol component through extraction.^[93] The liquid temperature range of some alloys such as eutectic gallium indium is ≈ 15 to 1300 °C.^[98] The large gap in air pressure between space and the ground also poses challenges. As a result of the ultrahigh vacuum in space, soft robots in Earth's orbit require structures such as fiber constraint layers to prevent channels from rupturing. On planets with low atmospheric pressure such as Mars, a soft robot can achieve the same deformation as that on Earth with a smaller pressure input, which facilitates a decrease in energy consumption. The performance of fluid-actuated soft robots is exactly the opposite on planets with thicker atmospheres. This implies stricter requirements for energy supply and structural strength. In addition, liquids can provide a higher output force than gases due to their low compression.^[158] In high-resistance environments such as the lunar surface, hydraulic actuation is often preferable to pneumatic actuation.

5.1.2. Smart Material Actuation

In space missions, lightweight and a large range of motion typically result in stronger operating capabilities. To overcome the dilemma of fluid-actuated soft robots carrying external pumps, soft robots can be actuated using other energy sources such as electricity, heat, and light.^[159,160] As a novel type of functional material, smart materials can sense external stimuli, discern, and respond to their environment. Smart material actuators have been used to actuate soft robots to perform tasks such as flight,^[17] crawling,^[27] capturing,^[16] and transportation,^[161] indicating strong potential for further development. For example, soft structures fabricated from thermally responsive smart materials have been used to deploy antennas in space.^[162,163] Regardless, to apply additional soft robots in space missions, actuators must be changed. Generally, soft robots based on electrically responsive smart materials exhibit high stability. Ozaki et al. developed a piezoelectric bending actuator.^[32] An insect-scale flapping-wing aerial vehicle actuated by such actuators can realize untethered controlled flight (Figure 5e). In another study, a flapping-wing aerial vehicle with soft multilayer DE actuators was demonstrated to withstand in-flight collisions without sustaining damage.^[20,21] Based on a curved unimorph piezoelectric actuator, the soft robot developed by Wu et al. can carry, climb, and manipulate its trajectory.^[164,165] Chung et al. introduced a tensegrity robot driven

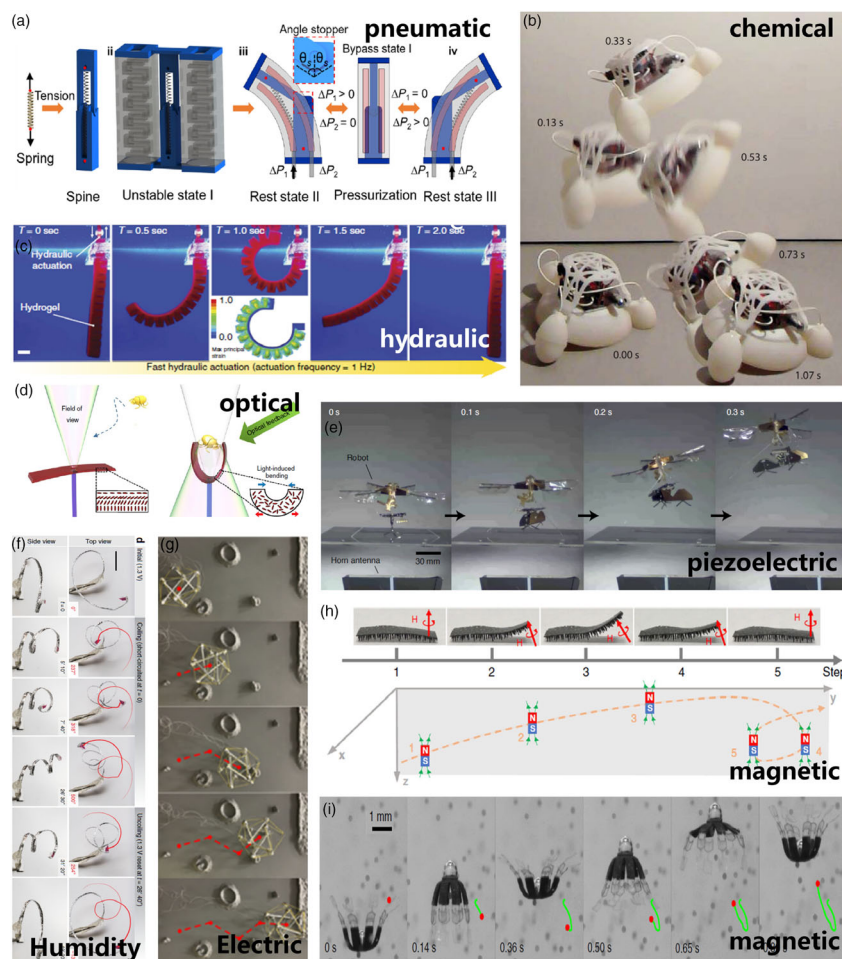


Figure 5. Typical actuation methods of soft robots. a) The schematic illustration of spine-inspired bistable soft actuators under both nonactuated and actuated states. Reproduced under the terms of the CC-BY-4.0 license.^[152] Copyright 2020, The Authors. Published by American Association for the Advancement of Science. b) A soft robot powered by mixed combustion of butane and oxygen. Reproduced with permission.^[26] Copyright 2015, Published by American Association for the Advancement of Science. c) A hydraulic soft manipulator. Reproduced under the terms of the CC-BY-4.0 license.^[154] Copyright 2017, The Authors. Published by Springer Nature. d) A light-driven artificial flytrap. Reproduced under the terms of the CC-BY-4.0 license.^[166] Copyright 2017, The Authors. Published by Springer Nature. e) Controlled flight of a flapping-wing aerial vehicle based on piezoelectric actuators. Reproduced under the terms of the CC-BY-4.0 license.^[32] Copyright 2021, The Authors. Published by Springer Nature. f) A tendril-like soft robot based on osmotic actuation. Reproduced under the terms of the CC-BY-4.0 license.^[169] Copyright 2019, The Authors. Published by Springer Nature. g) A tensegrity robot driven by torsional SMA springs. Reproduced with permission.^[33] Copyright 2019, American Chemical Society. h) A magnetic actuated soft robot. Reproduced under the terms of the CC-BY-4.0 license.^[167] Copyright 2018, The Authors. Published by American Association for the Advancement of Science. i) Locomotion of bionic jellyfish soft robot made of magnetic composite elastomer. Reproduced under the terms of the CC-BY-4.0 license.^[168] Copyright 2019, The Authors. Published by Springer Nature.

by torsionally prestrained shape memory alloy springs (Figure 5g).^[33] This robot can overcome obstacles by rolling or jumping. Because light is an energy source that can be controlled remotely, instantly, and precisely, light-actuated soft robots have significant space application prospects, particularly in deep space with stable light fields. Based on light-responsive liquid crystal elastomers, Wani et al. developed a bionic flytrap that is promising for space sampling (Figure 5c).^[166] It uses optical feedback to trigger photomechanical actuation for capture. Lu et al. reported a magnetically actuated soft robot with a multiple-tapered soft-foot architecture (Figure 5h).^[167] The robot can crawl, carry, and move over obstacles in both dry and wet sophisticated terrains. The bionic jellyfish soft robot proposed by Ren et al.

was fabricated using magnetic composite elastomers (Figure 5i).^[168] This robot can swim under an external oscillating magnetic field. Multimodal locomotion makes these soft robots applicable to the exploration of planetary surfaces and even Titan-like planets with lakes. However, in Earth's radiation belts, magnetically actuated soft robots may be difficult to operate accurately. Must et al. developed a tendril-like soft robot based on reversible osmotic actuation (Figure 5f).^[169] This robot has reversible rigidity and large-angle deformability, which may be useful for winding capture. Smart material actuators responding to external fields can realize the remote control of a soft robot, but the reliability of such systems is easily affected by the external environment. Overall, a protective design is indispensable.

The key is to select the appropriate actuation methods and materials based on the space mission.

5.2. Chemical Actuation

Chemically actuated soft robots use chemical reactions to convert the chemical energy of fuels into mechanical energy to perform movement. In general, chemical actuation can achieve large instantaneous output forces. Bartlett et al. developed a soft robot powered by the mixed combustion of butane and oxygen (Figure 5b).^[26] This robot can freely jump and fall smoothly. Jumping or rolling is often more efficient than crawling on a planetary surface with loose soil. In addition, some chemically actuated soft robots may adapt to special environments, such as acid or alkaline lakes found on some planets.^[170–172] Satoshi et al. successfully controlled the swelling and deswelling self-oscillation cycle of a polymer gel.^[173] This demonstrates that a chemically actuated soft robot can move stably in an acidic solution.

Table 3 summarizes the limitations and improvements of various actuation methods. Key data on the performance and space application potential of soft robots are presented in **Figure 6** and **Table 4**. We found that the deformability of soft robots is greater than that of rigid space robots at various scales (Figure 6a). As shown in Figure 6b, there is a significant negative scaling law between the relative speed and body mass of mammals, arthropods, rigid robots, and soft robots. Compared with the other three types, the distribution range of soft robots is wider, which significantly compensates for the deficiencies of rigid robots. We believe that this is related to the more diversified actuation methods of soft robots. In general, the force density of soft robots decreases with an increase in volume. Fluid actuators tend to exhibit the highest force density on the same scale (Figure 6c). The actuation performances of rigid space robots

and soft robots with space application potential are summarized in Table 4. Soft robots exhibit superior performance in many space applications.

6. Sensing and Control

Sensing and control are the core of effective interactions between soft robots and their environment. However, based on the large deformation of soft materials, the application of traditional electronic components in soft robots is limited in many aspects, including the inability to stretch. The harsh space environments impose stricter requirements on the reliability of sensors. Soft space robots require lighter, stronger, and more flexible sensors.^[202] Furthermore, the strong nonlinearity of soft materials presents significant challenges to the precise control of soft robot behavior.

6.1. Sensing

For soft space robots, it is essential to obtain multidimensional state information rapidly and accurately, which is inseparable from the support of sensors. Space equipment should meet three major features to ensure a pivotal role in future space systems, namely, adaptivity, multifunctionality, and self-healing.^[91] Compared with traditional sensors, flexible sensors have several advantages, including lightness, low energy consumption, and adjustable mechanical properties, which make them easier to integrate in soft space robots. Flexible sensors based on nanomaterials have been widely used to measure pressure,^[203] torsional force, shear force,^[204] temperature,^[205] strain,^[119] light pressure,^[206,207] capacitance,^[208] and a combination of these parameters. Space robots with integrated flexible sensors may significantly increase the payload efficiency of spacecraft.

Table 3. Space application limitations and possible improvement approaches for existing soft robot actuation methods.

Actuation type	Main limitation factor	Improvement approach	Suitable scenarios
Pneumatic	High vacuum, extreme temperature, microgravity	Temperature-resistant elastomers, thermal insulation system, radioresistance layer	On-orbit servicing, sampling, or cruise on planetary surface
Hydraulic	Extreme temperature, high vacuum, microgravity	Temperature-resistant elastomers, thermal insulation system, radioresistance layer	On-orbit servicing, sampling, or assembly
Magnetic	High-energy radiation, Earth radiation belts	Radioresistance layers/coatings	Remote operation (scenarios with a stable magnetic field)
Optical	Direct sunlight	Thermal insulation system	Remote operation (scenarios with a stable light field)
Piezoelectric	High-energy radiation, extreme temperature	Radioresistance layers/coatings	Scenarios of delicate manipulation or high-frequency locomotion
Electric (SMA/SMP)	Extreme temperature	Thermal insulation system, radioresistance layers/coatings	Remote operation (scenarios with a stable temperature field)
Humidity	Extreme temperature, high vacuum	Thermal insulation system, radioresistance layers/coatings	Participates in the humidity balance of the spacecraft
Chemical	Extreme temperature, high-energy radiation	Anticorrosion coatings, radioresistance layers/coatings	Highly acidic or alkaline environments, scenarios requiring large output

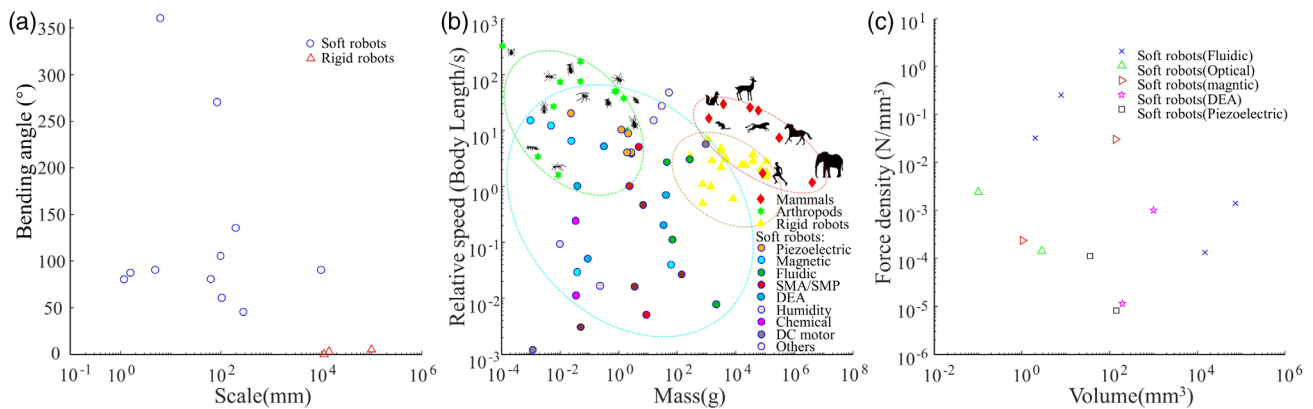


Figure 6. Deformability, locomotion, and power density of some soft robots and actuators. a) Bending angle of some soft robots and rigid robots versus their scale. b) Relative speeds of some mammals, arthropods, soft robots/actuators, and rigid robots versus body mass. c) Output force density versus volume of some soft robots and rigid robots. Data come from the studies.^[16,20,21,23,26,27,123,128,140,152,154,164,166,167,169,174–188] For Supplementary Data, see Table S1–S3, Supporting Information.

Table 4. Various actuation types for specific space applications and their performances.

Space missions		Actuation requirements	Performance of rigid robots	Performance of soft robots	Suitable types of soft robots
On-orbit Missions	Transfer	Load capacity	Over 3000 kg/8 times of own weight, 260 000 kg in gravityless conditions ^[189]	Over 400 times own weight ^[190]	Soft gripper Soft crawling robot
	Assembly	DOFs	6 to 23 ^[191]	Theoretically Infinite	Soft manipulator Electric-/Light-actuated soft robots
		Accuracy	1 mm (0.05% of the robot length) ^[192]	19.6 mm (2.23% of the robot length) ^[193]	
	Capture	Maximum relative speed ^{a)}	2 mm s ^{−1} ^[192]	2.5 m s ^{−1} ^[194]	Soft gripper Foldable soft robot
		Effective capture distance	0.2 m ^[192]	200 m ^[86]	
Planetary Exploration	Sampling	Drilling depth	Up to 2 to 3 m ^[195,196]	35 cm in sand ^[22] No report	Soft gripper Tip-growing soft robot
		Sampling mass ^{b)}	≈1731 g ^[197]		
	Detection	Payload mass	75 kg ^[198]	6 kg ^[199] on the ground	Soft crawling robot
		Passive deformability	No	Yes	Tip-growing soft robot
	Traverse	Distance traveled	>44 km ^[198]	>36 m on the ground ^[27] ≈691 m in water ^[200]	Flapping-wing air vehicle, soft crawling/rolling/soft jumping robot
		Maximum traverse speed	5 cm s ^{−1} ^[198]	18.75 cm s ^{−1} on the ground ^[152] ≈5 cm s ^{−1} on an uneven terrain ^[201]	

^{a)}A shaded background denotes better performance. The maximum relative speed refers to the speed of the capture mechanism relative to the target when it is ready for contact; ^{b)}The sampling mass refers to the mass of lunar or Martian soil sampled by space robots. The data here come from the Chang'e 5 mission.

The space-unstructured environment imposes strict requirements on materials, structures, functions, and applicable scenarios of flexible sensors. First, flexible sensors integrated in soft space robots require good shape adaptability. Stretchability in a specific direction can be achieved by constructing a mesh structure,^[209] horseshoe shape,^[210] wavy patterns,^[211] pop-up buckles,^[212,213] and fiber networks (Figure 7a–d).^[214,215] The energy supply of the sensors used in-space environments often

determines their service life. Pu et al. reported a soft skin-like pressure sensor (Figure 7g).^[216] This sensor can be closely attached to the skin or soft robotic surface and harvest energy from motion, thereby providing new opportunities for the self-energy supply of soft devices. External space equipment is vulnerable to impact from space debris and particles. Xun et al. used self-healing conductive polyurethane elastomers to develop a self-powered electronic skin (Figure 7i).^[110] The full

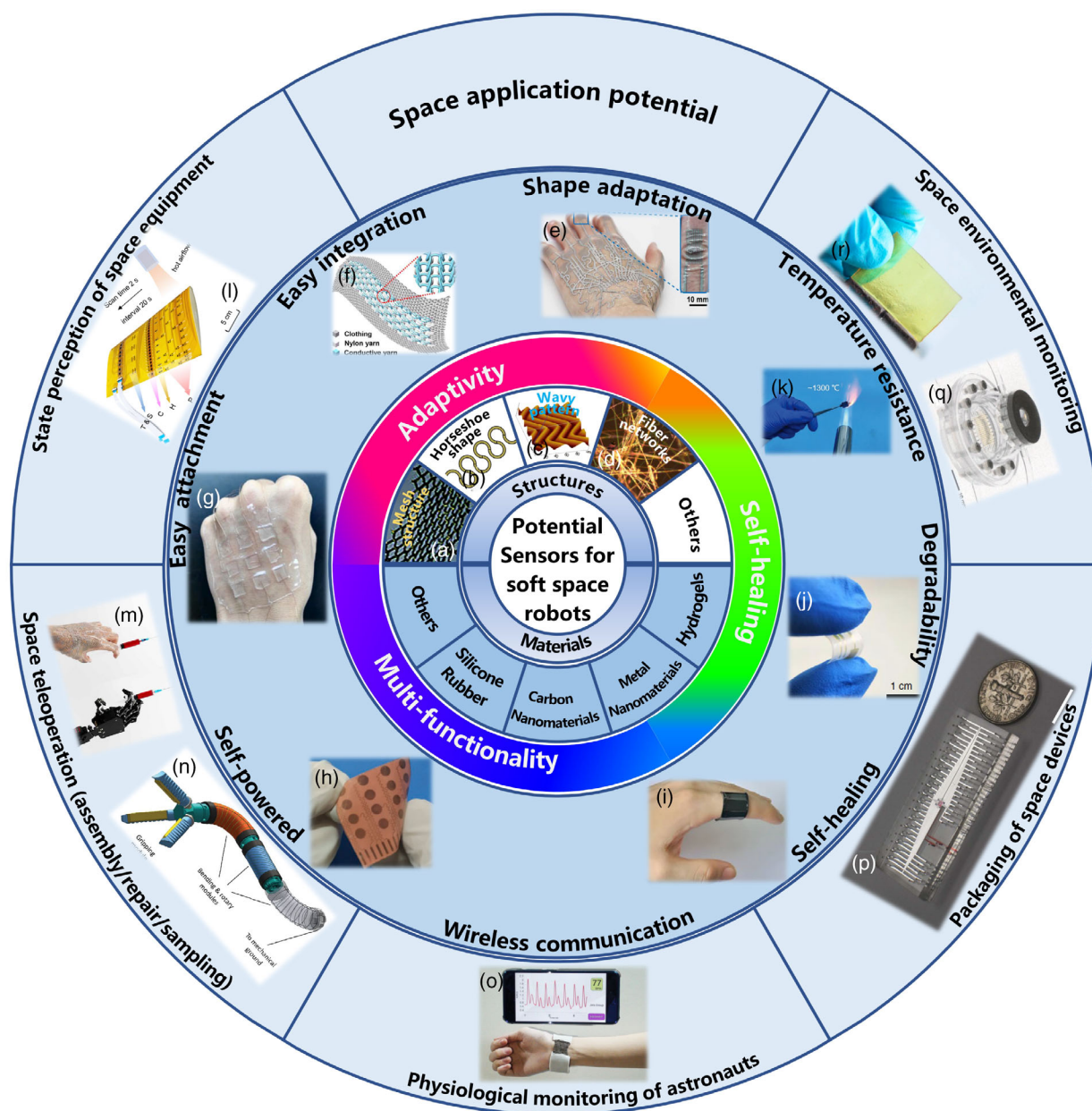


Figure 7. Structure, materials, functions, and application potential of flexible sensors for soft space robots. a–d) Flexible sensors constructed based on a mesh structure,^[209] horseshoe shape,^[210] wavy patterns,^[211] and fiber networks.^[215] Reproduced with permission.^[209] Copyright 2005, the National Academy of Science. Reproduced with permission.^[210] Copyright 2008, Elsevier. Reproduced with permission.^[211] Copyright 2007, American Chemical Society. Reproduced under the terms of the CC-BY-4.0 license.^[215] Copyright 2020, The Authors. Published by Springer Nature. e–k) Flexible sensors should satisfy some of the typical functions, such as shape adaption,^[224] easy integration,^[30] easy attachment,^[216] self-powered,^[110,218] wireless communication,^[218] self-healing,^[110] degradability,^[219] and temperature resistance,^[217] to accommodate space environments. Reproduced with permission.^[224] Copyright 2022, American Chemical Society. Reproduced under the terms of the CC-BY-4.0 license.^[30] Copyright 2020, the authors. Published by American Association for the Advancement of Science. Reproduced under the terms of the CC-BY-4.0 license.^[216] Copyright 2017, the authors. Published by American Association for the Advancement of Science. Reproduced with permission.^[110] Copyright 2020, American Chemical Society. Reproduced with permission.^[218] Copyright 2020, Elsevier. Reproduced under the terms of the CC-BY-4.0 license.^[217] Copyright 2020, the authors. Wiley-VCH. Reproduced under the terms of the CC-BY-4.0 license.^[219] Copyright 2020, The Authors. Published by Springer Nature. l–s) Currently, specific flexible sensors are expected to play a vital role in space applications, such as state perception,^[220] environmental monitoring,^[224,225] packaging,^[221] physiological monitoring,^[30] and teleoperation.^[222,223] Reproduced with permission.^[220] Copyright 2021, Elsevier. Reproduced with permission.^[224] Copyright 2022, American Chemical Society. Reproduced under the terms of the CC-BY-4.0 license.^[225] Copyright 2011, The Authors. Published by the National Academy of Sciences. Reproduced under the terms of the CC-BY-4.0 license.^[221] Copyright 2013, the authors. Published by Springer Nature. Reproduced under the terms of the CC-BY-4.0 license.^[222] Copyright 2021, the authors. Published by American Association for the Advancement of Science. Reproduced under the terms of the CC-BY-4.0 license.^[223] Copyright 2018, the authors. Published by Springer Nature.

self-healing capability exhibited by these sensors would lie at heart of extending space robots service life.^[91] To prevent circuit failure caused by extreme temperatures, flexible sensors should have a certain temperature resistance. The ceramic nanofiber sensor developed by Fu et al. can function at temperatures up to 1300 °C (Figure 7k).^[217] In another study, Guan et al. developed an electronic skin that can be extended to battery-free, wirelessly controlled, neural-stimulating robot systems (Figure 7h).^[218] In addition, the particularity of space equipment makes it possible to form space debris and pollute the space environment after decommissioning. Degradable flexible sensors are expected to alleviate this problem (Figure 7j).^[219] The above sensors usually have the ability to perform different functions, which is meaningful for reducing the mass and energy consumption of space robots.

The powerful functionality of flexible sensors provides several possibilities for space applications. The lightweight, large-scale electronic skin developed by Xiong et al. may endow space equipment with the capacity to sense, evaluate, and even predict flight states (Figure 7l).^[220] Similarly, space environment monitoring can effectively ensure the safety of on-orbit spacecraft and provide a basis for the protection design of spacecraft. Flexible sensors have been used to monitor exogenous pollutants (Figure 7r).^[224] and construct imaging systems (Figure 7q).^[225] However, because sensors for environmental monitoring are often exposed to outer space, effectively protecting them without affecting their performance remains challenging. Embedded flexible packaging based on temperature-resistant and radiation-resistant soft materials is one possible solution (Figure 7p).^[221] Space teleoperation is typical way to avoid direct damage to human body in harsh environments. In emergencies, this approach is beneficial for directly transferring human intelligence to robots. Tang et al. developed a highly stretchable, conformal, and sticky electronic tattoo (Figure 7e).^[224] Based on this sensor, soft robots can replicate the movements of a human hand in real-time during teleoperation (Figure 7m).^[222] Phillips et al. developed a glove-controlled soft robot (Figure 7n). The glove was embedded with flexible sensors to translate hand motions into soft robot actions to achieve deep-sea operations. Flexible sensors embedded in space suits enable the continuous monitoring of astronaut physiology. A machine-knitted washable sensor array (Figure 7f) developed by Fan et al. can monitor both pulse and respiration signals (Figure 7o).^[30]

6.2. Control

Controlling soft space robots with precision is a challenge based on the complexity of the space environment and high dimensionality of soft robots. Although controls for rigid robots are relatively mature, they are not applicable to soft robots with theoretically infinite DOFs.^[226] Therefore, the application of traditional control methods to soft robots is limited.^[227] In a space environment, microgravity and high vacuum can significantly affect the locomotion of soft space robots because extremely small forces can also cause significant changes in the motion state.^[35] This leads to control difficulties, particularly for high-speed dynamic control. In addition, extreme temperatures and high-energy radiation can significantly affect the stability of robot

control systems. All of these factors make the control of soft space robots more challenging. The control of soft robots can be classified into model-based and model-free methods depending on whether the control system contains a mathematical model of the controlled process.

6.2.1. Model-Based Control Methods

Most current applications of soft robots utilize static or quasidynamic controllers based on kinematic models.^[228] The PCC model is one of the most useful tools based on its simplicity and practicality.^[229] In the PCC assumptions, the dynamic effects of the end effector are frequently ignored. This assumption is more applicable in a space microgravity environment. The PCC model exhibits superior control performance for fluid-actuated and cable-actuated soft actuators.^[63,230,231] A soft robot for underwater operations verified the feasibility of this model in a microgravity environment.^[232] However, when environmental loads are significant or act on the nonconstant sections of a soft manipulator, control based on the PCC model often fails.^[65] Therefore, several nonconstant curvature models have been developed, including the spring-damper model^[233] and Cosserat approach.^[234] Renda et al. proposed a model for a multisegment soft manipulator based on the Cosserat approach.^[234] This model considers shear and torsional deformations, which are essential for coping with external loads in space. The mass-spring model has achieved promising results in the dynamic simulation of a tethered net robot for space debris capture.^[70,235] For soft robots with complex structures and strong nonlinearity, machine learning provides an alternative modeling method.^[238,237] Johnson et al. proposed a hybrid modeling approach based on machine learning that significantly improved the overall performance of a nonlinear predictive controller.^[236] The forward and inverse dynamic methods provide simple solutions. Thuruethel et al. used a recurrent neural network to represent the forward kinetic model for the closed-loop dynamic control of soft manipulators.^[228] In addition, it should be emphasized that, with adequate protection, extreme temperature and space radiation have limited impact on the modeling of soft space robots.

6.2.2. Model-Free Control Methods

Owing to the multivariable, strong coupling, and nonlinear nature of soft space robots, it is occasionally difficult to establish an accurate mathematical model of a system.^[236] Model-free control methods can be applied to multitype soft space robots, providing a wider application potential in unstructured environments.^[238] In complex space exploration missions, soft robots must overcome uncertainty and make timely decisions.^[239] In these scenarios, model-free control methods tend to perform better than model-based control methods. Some researchers have compared model-based and model-free soft robot control methods. The results demonstrate that model-free control is more accurate and efficient than model-based control in some cases.^[238,240] For example, for a tendon-driven soft robot for the maintenance of aeroengine combustors, Ba et al. developed a local model-free controller utilizing a fuzzy logic algorithm.^[241]

which effectively avoids mismatch and coupling problems compared with the PCC model. Giorelli et al. proposed a supervised learning method for solving the inverse static problem of nonconstant-curvature soft manipulators.^[242] The experimental results demonstrated the possibility of soft robot control in a microgravity environment. In addition, a behavior-based control system that integrates human-machine interfaces, such as electronic skins, can easily realize the intuitive space teleoperation of soft robots.^[243,244]

7. Potential Space Applications of Soft Robots

7.1. Soft Robots for On-Orbit Missions

Satellites, space stations, and space shuttles are typical orbiters that perform many tasks (e.g., navigation, communication, reconnaissance, and scientific experiments). Typically, orbiters carry space robots to assist in complex tasks. Potential space missions, corresponding robotic system performance requirements, and candidate types are listed in Table 5. Soft robots have significant potential based on their high flexibility and robustness and have inspired many new designs. Considering the difficulty and cost of rocket launching, space-deployable devices, such as space antennas and solar cells, are conducive to alleviating the contradiction between their weight and effective utilization area.^[245] Therefore, a soft space foldable mechanism with lighter weight and more compact storage will be more suitable for future spacecraft.^[246] Ikeya et al. proposed a multifunctional deployable

membrane based on a bisymmetric packaging scheme that reduces volume through folding and crimping (Figure 8a).^[10] This method can be applied to the deployment of solar cells and array antennas. Kirigami art provides a novel solution for realizing the deformation of flexible space structures. The dynamic kirigami structures developed by Lamoureux et al. enabled solar cells to obtain more efficient optical tracking through the tilt generated by stretching (Figure 8b).^[24] Inflatable release is another rapid deployment method for space-deployable structures. The inflatable antenna experiment led by NASA adopted Z-fold cylindrical arms as antenna booms. These 28 m-long soft folding arms can be deployed rapidly after inflation (Figure 8e).^[34,247,248] On-orbit inspection and maintenance are extremely complex tasks because they often require extravehicular operations.^[5] Due to the complexity of such tasks, space robots for on-orbit maintenance require high-space DOFs and damage resistance. Hawks et al. reported a soft pneumatic robot that realized rapid and significant lengthening of its body through pressurization and inflation.^[13] In addition, the robot can move through passive deformation in constrained environments.^[13] Such robots are expected to be used for space station maintenance and planetary exploration (Figure 8c).^[249] In another study, a robot developed by Blumenschein et al.^[14] was used to construct 3D structures such as space radio antennas by lengthening its preformed body. This robot can also capture objects via helical actuation, which may be useful for satellite capture. Araromi et al. proposed a novel method for active debris removal.^[16] Their soft gripper using rollable DE structures can

Table 5. Various space missions and their performance requirements for robots.

Space missions		Performance requirements for robotic system	Types of candidate robots
On-orbit missions	Mount antenna	Payload mass ≈ 80 g ^[252]	Rigid space robots Inflatable soft robots
		End effector accuracy $\approx \pm 2$ mm ^[252]	
		Allowable time delay ≈ 0.5 s ^[253]	
	Assemble solar arrays	Payload mass ≈ 100 g ^[252]	Rigid space manipulator Multiple mobile arm space robots ^[253]
		End effector accuracy $\approx \pm 2$ mm ^[252]	
		Maximum apply force ≈ 20 N ^[252]	
		End effector opening ≈ 1 mm ^[252]	
		Allowable time delay ≈ 0.5 s ^[253]	
	Replace optics	Standard camera mass < 1 kg ^[252]	Rigid space manipulator
		End effector accuracy $\approx \pm 1$ mm ^[252]	
	Replace micropropulsion system	Payload mass < 5 kg	Rigid space manipulator
		End effector opening ≈ 80 mm	
Planetary exploration	Remove broken subsystem	6-DOF giving high dexterity ^[254]	Rigid space manipulator
		Applied force < 10 N ^[35]	
	Space docking	Payload mass < 10 kg ^[255]	Rigid space manipulator Net capturing robot ^[235]
		End effector opening ≈ 10 cm ^[255]	
	Remove of small space debris	Reach of ≈ 1 m ^[256]	Soft tether-gripper robot ^[235] Soft continuum manipulator ^[257]
		Responding speed ≈ 5 cm s ⁻¹ ^[255]	
		Maximum sampling mass (robot) ≈ 2 kg ^[197]	Rigid robot manipulator, soft gripper, Pneumatic soft robot
	Soil sampling	Sampling depth of up to ≈ 2 –3 m ^[195,196]	
		Distance traverse ≈ 0.1 –44 km ^[198]	
	Traverse	Maximum traverse speed ≈ 0.6 –5 cm s ⁻¹ ^[198]	Mars rover, Lunar rover, soft crawling robot
	Establish habitat	Formable Lunar regolith cover > 2 m ^[258]	Multiple mobile arm space robots, ^[259] cable-driven robot ^[146]
		6-DOF giving high dexterity ^[146]	

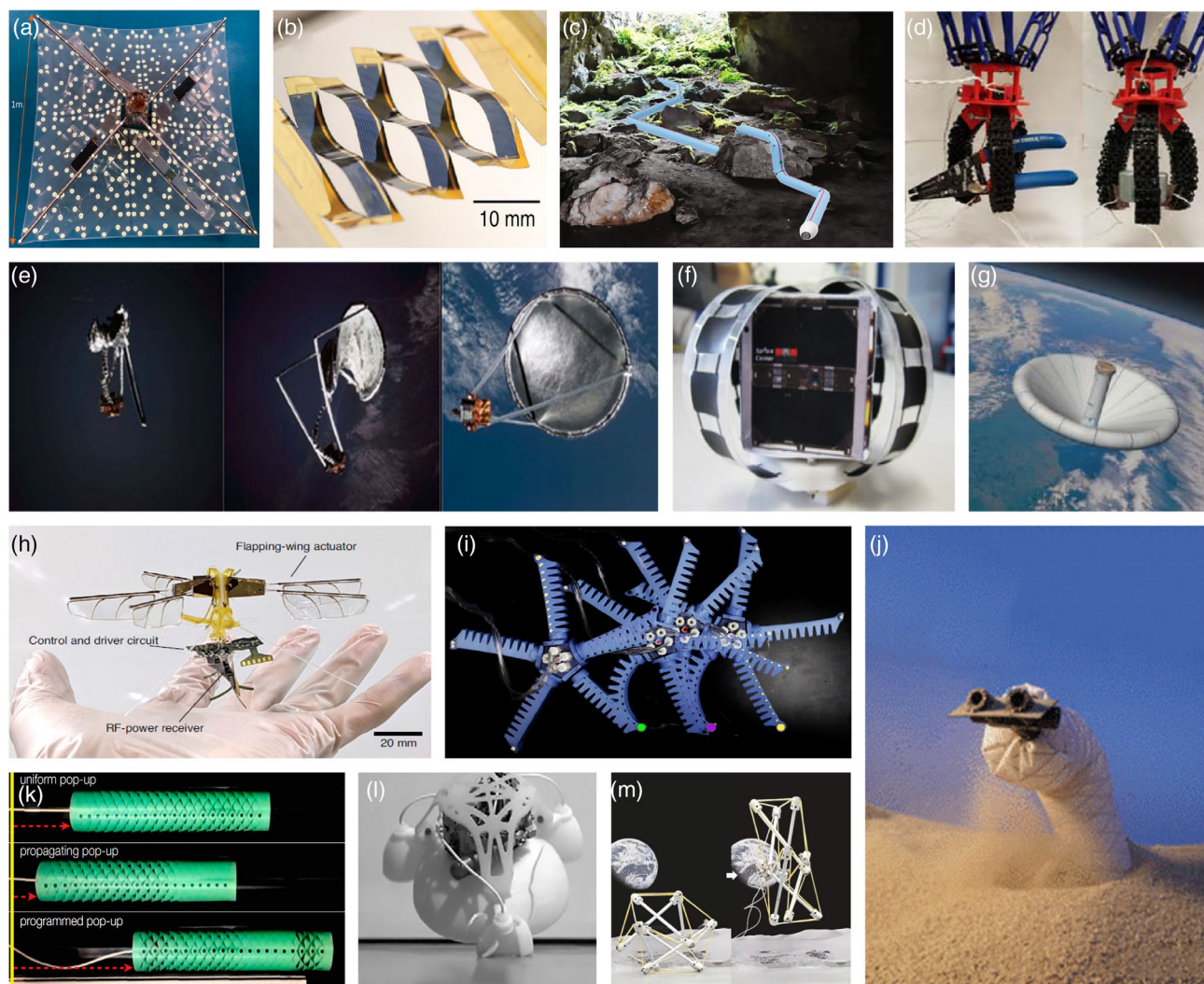


Figure 8. Soft robots with potential space applications. a) Origami structure for solar arrays. Reproduced with permission.^[10] Copyright 2020, Elsevier. b) Solar cells for solar tracking based on kirigami structures. Reproduced under the terms of the CC-BY-4.0 license.^[24] Copyright 2015, the authors. Published by Springer Nature. c) Soft robot that navigates through growth. Reproduced under the terms of the CC-BY-4.0 license.^[249] Copyright 2021, the authors. Published by Science and Technology Review Publishing House. d) Soft gripper with pressure sensing. Reproduced under the terms of the CC-BY-4.0 license.^[29] Copyright 2019, the authors. Published by Wiley-VCH. e) Space-deployable inflatable antenna. Reproduced with permission.^[34] Copyright 2014, American Institute of Aeronautics and Astronautics. f) Soft gripper for satellite capture. Reproduced with permission.^[16] Copyright 2015, IEEE. g) Soft inflatable decelerator based on aerogel composite materials. Reproduced with permission.^[251] Copyright 2011, American Chemical Society. h) Wireless microscale flapping-wing aerial vehicle. Reproduced under the terms of the CC-BY-4.0 license.^[32] Copyright 2021, the authors. Published by Springer Nature. i) Starfish-inspired soft robot. Reproduced under the terms of the CC-BY-4.0 license.^[269] Copyright 2020, the authors. Published by Wiley-VCH. j) Steerable, burrowing root-like soft robot. Reproduced with permission.^[22] Copyright 2021, American Association for the Advancement of Science. k) Kirigami-inspired soft robot. Reproduced under the terms of the CC-BY-4.0 license.^[270] Copyright 2019, the authors. Published by the National Academy of Sciences. l) Combustion-powered jumping robot. Reproduced with permission.^[26] Copyright 2015, American Association for the Advancement of Science. m) Tensegrity robot capable of rolling and jumping. Reproduced with permission.^[33] Copyright 2019, American Chemical Society.

capture objects of different shapes and sizes while exhibiting good consistency (Figure 8f). Overall, softer robotic grippers exhibit better adaptability to complex targets. Teoh et al. developed a folding polyhedron that can be used to capture marine organisms.^[250] Furthermore, the robot is also desirable for space applications, such as alien deployable habitats and solar arrays. Kaur et al. reported a soft gripper that integrated embedded

pressure sensors (Figure 8d),^[29] which provides the possibility for the closed-loop space operation of soft robots. Yang et al. presented a variable-stiffness flexible robot for noncooperative target capture in space.^[193] This SMA-spring-actuated soft robot exhibited excellent system robustness. The ratio of the positioning error to the robot length was less than 2.23% at the end effector. In addition, the light and flame-retardant performance of

aerogels make them promising candidates for thermal protective equipment such as spacesuits. Thus far, soft inflatable decelerators based on aerogel composite materials have successfully protected the landings of several Martian landers (Figure 8g).^[251]

7.2. Soft Robots for Planetary Exploration

Planetary exploration is one of the most challenging tasks in the field of space exploration. In recent years, an increasing number of countries have listed exploration of the Moon and Mars as the next phase of space programs. Currently, all launched planet probes are rigid robots. However, due to the restrictions of the harsh environments of extraterrestrial planets, rigid robots with limited freedom are prone to faults. For example, the Opportunity rover failed because its solar cells were covered with dust generated by dust storms.^[260,261] Therefore, soft robots with better flexibility and robustness have significant potential for planetary exploration. The controlled flight of Ingenuity on the surface of Mars proves the feasibility of unmanned aerial vehicles for Mars exploration. Compared to rotary-wing and fixed-wing aircraft, flapping-wing aircraft are less sensitive to low Reynolds numbers,^[18,19] which can play a role in Mars dust detection to understand its global climate.^[262] Inspired by the flight of bumblebees, the microscale light-powered flapping-wing aerial vehicle developed by Jafferis et al. achieved sustained unrestricted flight.^[17] In another study, Ozaki et al. successfully actuated an insect-scale flapping-wing aerial vehicle with a radio-frequency power supply (Figure 8h).^[32] The high mobility and robustness of such aircraft indicate the possibility of environmental monitoring and navigation in a restricted environment on a planetary surface.^[17,20,32] In highly unstructured environments, soft robots can explore using structural deformation. NASA has developed a pop-up flat-foldable robot for planetary surface detection.^[15] This robot can climb slopes or cross caves through folding deformation, providing significant mobility advantages. Wu et al. proposed a soft robot based on a curved unimorph piezoelectric structure.^[164] The relative speed of the robot can reach 20 body lengths per second. In addition, it can bear a weight that is one million times heavier its own weight, demonstrating strong mobility and robustness. This robot has potential applications in environmental detection, information reconnaissance, and structural inspections. Jumping is an efficient method for moving on the surfaces of planets with low gravity and loose soil.^[263–265] Bartlett et al. fabricated a combustion-actuated robot that could jump freely and land smoothly (Figure 8l).^[26] Hawkes et al. developed a jumping robot that can jump 125 m high while covering 500 m in a single bound when on the Moon.^[265] A soft body provides a robot with strong impact resistance, which is difficult to achieve for a rigid space robot. The granular soil on some planetary surfaces causes difficulties in the movement of conventional space robots.^[2] Naclerio et al. reported a steerable root-like tip-growth soft robot that could dig holes through sand and even avoid underground obstacles by controlling its direction of motion (Figure 8j).^[22] The realization of this robot provides a novel solution for extremely complex applications, such as soil sampling on the surface of planets or asteroids and volcanic tunnel exploration of the Moon. On Mars, the gravity is lower but winds are much faster

than Earth's surface.^[266,267] The conventional Mars rovers are easily damaged. One of the possible solutions may refer to the zoobenthos, such as echinoderms, pelecypods, and reptiles. The buoyancy and turbulent environment shape their special morphology and gaits so they can move along the seabed.^[268] Scott et al. developed a starfish-inspired soft robot and found optimally effective gaits of the robot through planar discrete elastic rod theory modeling (Figure 8i).^[269] Branyan et al. created a soft robot inspired by snakes.^[174] This robot imitates the movement of a snake and navigates by changing the speed and amplitude of a traveling wave. This movement pattern enables the robot to move in different particle media environments ranging from fine gravel to stone. In another study, based on kirigami skins, a soft robot designed by Rafsanjani et al. effectively crawled on rough surfaces (Figure 8k).^[270] This research provided a promising skin design for planetary rovers used in space exploration.^[271] Tolly et al. developed a soft pneumatic untethered crawling robot.^[27] Soft composites make the robot inherently resistant to harsh environments, such as snow, flames, and even car rolling. These studies provide inspiration for the design of future cruising robots on the planetary surface. Another major obstacle restricting the space applications of soft robots is the electronic components in soft robot systems, which reduce the resistance of robots to space radiation and extreme temperatures. Drotman et al. proposed a soft-legged robot without electronics.^[272] This robot controls its gaits using only a simple pneumatic circuit. A soft spherical tensegrity robot is another type of soft robot that is clearly toward space exploration. Chung et al. developed a tensile robot capable of fast rolling and jumping with strong mobility and robustness on a surface simulating the Moon (Figure 8m).^[33] In addition, Usevitch et al. reported an inflatable truss-like robot with similar principles that also exhibited potential for exploring planets and carrying objects.^[161]

7.3. Soft Robots for Deep Space Exploration

In addition to space missions such as on-orbit operation and near-Earth planetary exploration, deep space exploration activities will become a key development field in the future. Soft structures have been used in this field. For example, aerogel is considered an ideal material for space equipment based on its excellent characteristics of light weight, heat insulation, and sound insulation.^[251] Aerogel composites have been used in Mars rovers,^[273] spacecraft parachutes,^[273] spacesuits, and even in space particle capture.^[274] However, aerogels have poor mechanical carrying capacities,^[275] which limit their further application.

8. Discussion

Soft robots have significant potential for space applications. Soft structures and robots for space applications have achieved impressive results. However, several key challenges must be addressed to obtain more practical soft space robots. 1) The impact mechanisms of the space environment on soft robots have to be comprehensively evaluated. Unlike in the ground environment, the impact mechanisms of the space environment on soft robots and their components are unclear. In this review, the

effects of microgravity, ultrahigh vacuum, strong radiation, and extreme temperature on soft robots were discussed. In a real space environment, many other factors exist. Previous laboratory tests typically considered only the impact of a single space environmental factor on a single material or device. Current ground testing conditions cannot simulate a complete set of space environments. Therefore, little is known about the combined effects of space environment on soft robots. 2) The design criteria for soft space robots are determined. Currently, the design theory for soft robots has not yet formed a general system. The specific requirements of various types of space missions are currently unclear. For example, the impact of space debris in different orbits on the environment is enormous. It is difficult for current rigid and soft robots to capture space debris to achieve the desired effect. A deployable gripper designed without preconditions has the potential to handle high speed, collision, and unstructured motions. Therefore, it is necessary to clarify the design criteria for soft space robots. 3) Actuators with higher performance should be developed. Based on the difficulty of maintaining space devices, soft space robots should have high actuation efficiency and reliability. The actuators of soft space robots should satisfy the requirements of high repeatability, robustness, and long-term stability and be capable of adapting to harsh space environments. 4) Fabrication technology for soft robots requires further progress. The fabrication technology for soft space robots should satisfy the requirements of rapid, low-cost, and integrated manufacturing processes. Functional facilities, including actuation, sensing, computing, and protection units, should be embedded in soft robots to ensure their adaptability in space environments and missions. 5) Improving modeling and control theory for soft space robots. Based on the inherent strong nonlinearity of soft materials, their modeling and control theory are not mature even for soft robots applied to the ground. In harsh space environments, the modeling and control theories of soft robots are more complex. Most current control methods for soft robots remain quite limited in their adaptation to highly unstructured space environments.

9. Conclusions and Prospects

In this article, we summarized major advancements in soft robotics and their potential space application in recent years from the perspectives of design and modeling, fabrication, actuation, sensing, and control. The special requirements for the design of soft robots for space applications were discussed by analyzing the characteristics of the space environment and space missions. Furthermore, we analyzed the current challenges and future tendencies of soft space robots, focusing on the aspects of soft robots with potential space applications.

In space engineering, the intelligence and integration of equipment have become important research trends. The development of soft robotics has reshaped the philosophy of robotic design, fabrication, and control. Soft robot technology, which incorporates several advanced technologies such as smart materials, flexible electronics, advanced actuation technology, and control theory, may promote the intelligence of space equipment. Soft structures have been successfully applied in space equipment (e.g., solar arrays, inflatable space station modules, and

satellite grippers). Research has confirmed the ability of soft robots to perform multimodal movements in complex terrain environments, delicate operations, and human–computer interactions. Compared with traditional space robots, soft robots have apparent advantages of flexibility, dexterity, robustness, and safety, and exhibit better adaptability to space environments. However, regarding modeling and control difficulties, the limitations of current soft robots are also noticeable.

It is important to determine how we can overcome the bottlenecks of soft robots in space applications in the future. First, it is necessary to deeply understand the space environment and clarify the comprehensive impacts of space environments on soft robots. This will allow us clarify the design criteria for soft space robots and fabricate soft robots that satisfy specific requirements. Currently, several key challenges exist, including actuators with higher actuation efficiency and long-term stability, low-cost integrated fabrication methods, and more accurate modeling and control theory. A highly unstructured space environment may amplify the mismatches between the mechanical characteristics of soft robots. Advanced materials are considered key to effectively solve this problem. Furthermore, soft robots must respond promptly to changes in their external environment and status. Functional facilities, including sensing, actuation, computing, and protection units, should be embedded in soft robots to endow them with autonomous perception, judgment, and movement. Therefore, these materials are expected to become increasingly similar to computers. Empowered by various technologies such as artificial intelligence, soft space robots can achieve the in-depth integration of perception, actuation, control, and fabrication. Soft space robots will aid in establishing multilevel connections between simple operations, high-level tasks, and human–computer interactions. Therefore, soft robots will become autonomous systems and serve the development of the aerospace industry.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

soft robotics, potential applications, space environments, space explorations

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